

Key Learnings From My Data Cleaning Project

Data cleaning is the process of finding and fixing issues in raw data so it becomes accurate, consistent, complete, and actually usable for analysis. I used to think the goal was to make the data "perfect," but after spending real hours on this, I realised the actual goal is simpler than that: make the data reliable enough to answer the questions you're asking of it.

Main Types of Data Cleaning

- Handling missing values
- Removing unintended duplicate records
- Correcting data types
- Standardizing text (capitalization, spelling, spaces, formats)
- Validating values against expected rules
- Investigating outliers where relevant

Recommended Workflow

- 1 Work through the dataset systematically, rather than jumping around.
- 2 For each issue (missing values, casing, invalid values, and so on), check the relevant columns one by one instead of trying to fix everything at once.
- 3 Prefer small, readable SQL queries over one very large and complex query. Clear, understandable SQL is more valuable than solving everything in a single statement.
- 4 Remove duplicate rows only when they are genuinely unintended duplicates, not just similar-looking records.
- 5 If recovering a missing value requires complex inference and the column isn't important for your analysis, it's fine to leave it as NULL.
- 6 If you aren't highly confident about an inferred value, don't guess. Leave it as NULL.
- 7 Document every cleaning decision and the reason behind it, so future-you (or anyone else) understands why a choice was made.

What This Looked Like in Practice

In practice, this workflow turned into a step-by-step SQL pipeline: fixing inconsistent casing in text fields, trimming trailing whitespace, filling in nulls where it made sense, and flagging duplicate records. None of these steps were individually hard. What took the hours was slowing down enough to check each column properly instead of assuming it was already clean.

A Few Things I'd Tell Myself Earlier

- Messy data rarely announces itself. A column can look fine at a glance and still have trailing spaces or inconsistent casing hiding inside it, so it's worth checking properly rather than trusting a first look.
- Writing five small, readable queries beats writing one clever query that does everything. The small ones are easier to debug, and easier to explain later.
- Leaving a value as NULL is a legitimate decision, not a failure. Guessing just to fill a gap creates a problem that's harder to spot later than an honest NULL.
- Duplicates are not always what they look like. Two similar rows might represent two real, different events, so it's worth checking before deleting anything.
- Writing down why I made each cleaning decision, while I was still in the middle of it, saved me from having to reverse-engineer my own logic later.

The Golden Rule

Correctness, clarity, and explainability matter more than overly clever SQL. It's fine to combine queries when it genuinely improves efficiency, but never at the cost of readability. If I can't explain a cleaning step in one sentence, it probably needs to be broken down further.

This is the version of the guide I wish I had before starting the project. Coming back to it before the next dataset.